SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

DOCUMENT CONTROL

Version	Date	Prepared By	Reviewed By	Approved By	Description of Changes
1.0	2026-08-15	Anjali Patel (Project Manager)	Vikram Desai (QA & Sec Lead)	Rajesh Sharma (VP E-Commerce)	Initial SRS baseline for Aanya Fashion
2.0	2026-10-06	Anjali Patel (Project Manager)	Rahul Mehta (Tech Lead)	Rajesh Sharma (VP E-Commerce)	Added Boutique POS, GST Tax Engine, and WebAssembly AI Specs
2.1	2026-10-12	Anjali Patel (Project Manager)	Rajesh Sharma (VP E-Commerce)	Rajesh Sharma (VP E-Commerce)	Formal PMO approval and operational go-live sign-off

Project Name: Aanya Fashion (Aanya Fashions Omnichannel Platform)

Document Version: 2.0 (Engineering Baseline)

Prepared By: Anjali Patel (Project Manager)

Date: October 06, 2026

1. INTRODUCTION

This Software Requirements Specification (SRS) establishes the complete functional, non-functional, and interface requirements for Project UT-FEC-2026-01: Aanya Fashion Omnichannel E-Commerce & Retail Platform. Managed by Project Manager Anjali Patel, the document serves as the formal baseline for engineering implementation, quality assurance, and stakeholder acceptance.

The platform bridges digital retail and brick-and-mortar boutique store operations into a unified cloud-native architecture, delivering synchronized inventory, high-performance web shopping, and boutique cashier point-of-sale billing.

1.1 PURPOSE

The purpose of this SRS is to define the exact technical and operational specifications governing the development, deployment, and validation of the Aanya Fashion platform.

This document provides an authoritative reference for project managers, software engineers, QA specialists, and retail business stakeholders, ensuring complete architectural alignment across all deliverable modules.

1.2 SCOPE

The scope encompasses the customer-facing web storefront (catalog exploration, AI search, cart, and

Indian payment checkout), administrative control portal (catalog CRUD, category management, order fulfillment, and customer CRM), and the in-store boutique POS cashier counter (SKU/barcode lookup, GST invoicing, and thermal receipt printing).

Also in scope are real-time atomic inventory synchronization via BroadcastChannel and database triggers, client-side WebAssembly photo background removal (@imgly/background-removal), and automated Instagram catalog syndication (@aanya.style). Phase 1 explicitly excludes native mobile binary applications and cross-border currency conversion.

1.3 DEFINITIONS, ACRONYMS & ABBREVIATIONS

Term	Definition
POS	Point of Sale - Boutique cashier counter billing terminal for in-store walk-in transactions
UPI	Unified Payments Interface - Real-time mobile payment system developed by NPCI in India
RLS	Row-Level Security - PostgreSQL engine policy mechanism restricting row access by user role
WASM	WebAssembly - High-performance binary code format enabling client-side browser neural inferences

2. FUNCTIONAL REQUIREMENTS

Req ID	Requirement Description	Priority	Source
FR-01	Storefront & Catalog Exploration: High-performance single page application with Bromello typography, dynamic category filtering (Sarees, Kurtis, Lehengas), image zoom, and phonetic/semantic AI search.	High	BRD-01 / Marketing
FR-02	Indian Payment & Checkout: Multi-step checkout supporting UPI intent & dynamic QR, RuPay/Cards, 50+ domestic net banking institutions, and Cash on Delivery (COD) with address validation.	Critical	BRD-02 / Finance
FR-03	Boutique In-Store POS Terminal: Fast cashier counter with USB barcode scanner & SKU lookup, bespoke alteration line items, multiple payment tenders, and atomic inventory deduction.	Critical	BRD-03 / Retail Ops
FR-04	GST Invoicing & Dual Printing: Automated CGST/SGST tax calculation, numberToWordsINR currency translation, and instant 58mm/80mm ESC/POS thermal receipt & A4 formal tax invoice generation.	High	BRD-04 / Accounting

Req ID	Requirement Description	Priority	Source
FR-05	Synchronized Real-Time Inventory: Instant multi-channel stock deduction upon in-store or online purchase with zero-delay BroadcastChannel messaging and low-stock reorder thresholds.	Critical	BRD-05 / Inventory
FR-06	AI Media & Instagram Syndication: Client-side WebAssembly background removal for catalog images and automated @aanya.style Instagram feed caching with canonical shortcode resolution.	Medium	BRD-06 / Creative

3. NON-FUNCTIONAL REQUIREMENTS

Category	Requirement	Acceptance Criteria
Performance	Sub-1.8s Largest Contentful Paint (LCP), POS transaction completion under 30 seconds, and sub-60 second online checkout.	Lighthouse performance score >= 90; load tested for 500 concurrent shoppers.
Security	Strict PostgreSQL RLS policies, TLS 1.3 enforced, pgcrypto password hashing, and DPDP Act compliance.	Zero critical vulnerabilities in OWASP Top 10 automated pen-tests; zero unencrypted PII.
Availability	99.9% platform availability with multi-zone cloud failover, CDN edge routing, and automated daily database backups.	Monthly unplanned downtime under 43 minutes; RTO < 15 minutes, RPO < 5 minutes.
Usability	Mobile-first responsive UX with luxury ethnic design, intuitive POS cashier shortcuts, and number-to-words tax display.	Boutique cashier onboarding completed in under 2 hours with zero operational errors.
Scalability	Elastic asset delivery on Vercel Global Edge CDN, Supabase database connection pooling, and optimized WebP media.	Seamless traffic throughput during festive flash sales (Diwali, Navratri, Wedding Season).

4. EXTERNAL INTERFACE REQUIREMENTS

User Interfaces: Responsive web interface optimized for modern desktop browsers (Chrome, Safari, Firefox, Edge) and mobile devices (iOS Safari, Android Chrome). Implements luxury Indian ethnic styling with Bromello typography, responsive layouts, accessible Radix UI primitives, and motion micro-interactions.

Hardware Interfaces: Boutique retail checkout terminals require support for standard USB/HID plug-and-play barcode scanners, 58mm and 80mm ESC/POS thermal receipt printers (via CSS @media print), and standard A4 laser printers for formal GST tax invoices.

Software & Cloud Interfaces: Supabase PostgreSQL cloud database (REST API, Realtime Broadcast, and JWT authentication), Indian domestic payment gateways (Razorpay/UPI deep links), client-side

WebAssembly neural execution runtime, and Instagram Graph API syndication pipeline.

5. ASSUMPTIONS & CONSTRAINTS

1. Regulatory Compliance: The platform operates in strict compliance with the Indian Digital Personal Data Protection (DPDP) Act and domestic GST invoicing mandates.
2. Currency & Regional Scope: All monetary transactions, invoicing, and financial accounting ledgers are executed exclusively in Indian Rupees (INR - ₹).
3. Hardware & Infrastructure: Boutique billing counters require standard desktop/laptop hardware with USB ports and stable broadband connectivity for real-time cloud inventory sync.

APPROVAL

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